AIMCR FORENSIC REPORT

AI-Powered Mobile Forensic Intelligence Platform for Autonomous Crime Reconstruction

Case Reference:	Smoke Test Case - E2E Validation
Case Status:	Active
Date Generated:	2026-07-21 20:42:33
Assigned Investigator:	Lead Investigator John
Evidence Count:	1 artifacts ingested
Forensic Complexity Index:	0/100 (Low)

Executive Summary

This document compiles explainable investigative findings extracted from mobile device artifacts using the Autonomous Crime Reconstruction and Behavioral Profiling engines. Every finding detailed within this report is traceably backed by a corresponding SHA-256 hashed forensic artifact, maintaining full chain of custody. The contents represent investigative directions and indicators of probability rather than definitive judicial outcomes.

1. Incident Reconstruction

No structured cybercrime sequences were reconstructed by the engine.

2. Behavioral Anomaly Log

The Isolation Forest engine analyzed the user's historical feature vectors (active hours, GPS coordinates, application categories, file extensions) and highlighted the following deviations from baseline behavior:

No behavioral deviations were flagged by the anomaly detector.

2a. Anti-Forensics & Tampering Log

Heuristic scanners inspected the integrity of the device artifacts to find evidence of active concealment, extension spoofing, file system timestamp manipulation, or database tampering:

No active anti-forensics attempts or database anomalies were flagged.

3. Relationship Network & Priority Entities

The communication graph maps contacts, browser domains, and apps into connected modular communities. Nodes are assigned an Investigative Priority score using degree and betweenness centralities.

Priority Investigation Targets

Entity Identifier	Node Class	Investigative Priority Score	Action Needed
+15550011	Contact	60.0/100	Monitor & Validate
SMS Messenger	Application	60.0/100	Monitor & Validate

3a. Forensic Investigation Recommendations

AI/Deterministic Recommendations Summary: Based on the analysis of active threat intelligence matches and anti-forensics indicators, the investigator should prioritize disassembling the spoofed APK binary (disguised as a PNG) and carving the tampered SQLite database. Reconstructing network infrastructure and verifying modification timestamps via system logs is critical to confirm the attack vector and trace perpetrator activity.

Recommended Actions	Priority	Forensic Rationale / Next Steps
Establish forensic write-block and compute device image hash	Medium	Ensures all evidence remains statically secure and complies with chain of custody regulations.
Verify device acquisition logs for cryptographic integrity matches	Medium	Verifies that no bit-level changes occurred during evidence ingestion.

4. Evidence Integrity Index (Chain of Custody)

Every artifact imported into the platform is indexed below with its original acquisition timestamp and a SHA-256 integrity hash to ensure transparency and auditability.

Evidence ID	Timestamp	Type	Source File	SHA-256 Hash Signature
fe4f94f9...	2026-07-20 14:30:00	SMS	test_evidence.json	9863dca150e1a386...

5. System Investigation Audit Trail

Immutable audit log records detailing investigator queries, report configurations, and analyses run inside this case file:

Timestamp	Action	Investigator details / Queries
2026-07-21 15:12:24	QUERY_COPILOT	Asked Copilot: 'What is this investigation about?'. Response cited 1 artifacts.

2026-07-21 15:12:16	RUN_ANALYSIS	Executed AI behavioral profiling & communication graph analysis for case ID: eda801c0-ea4f-47e8-9d88-6ba0b9c39524. Detected 0 anomalies and 0 Threat Intel match(es).
2026-07-21 15:12:16	UPLOAD_EVIDENCE	Uploaded forensic file 'test_evidence.json' (Type: SMS, Hash: 9863dca150e1...) to case ID: e696af56-7515-470a-87b0-9b3d6868c797
2026-07-21 15:12:16	CREATE_CASE	Created case 'Smoke Test Case - E2E Validation' (ID: e696af56-7515-470a-87b0-9b3d6868c797)
2026-07-21 15:12:16	LOGIN	User investigator@aimcr.gov logged in successfully
2026-07-21 15:07:45	RUN_ANALYSIS	Executed AI behavioral profiling & communication graph analysis for case ID: 2c045cb8-845b-4d0c-b22f-ab6b4aadf10e. Detected 4 anomalies and 4 Threat Intel match(es).
2026-07-21 15:07:45	LOGIN	User investigator@aimcr.gov logged in successfully
2026-07-21 15:04:34	LOGIN	User admin@aimcr.gov logged in successfully
2026-07-21 15:04:34	EXPORT_REPORT	Generated PDF Investigation Report for Case 'Operation phoenix' (Path: C:\Users\Admin\Documents\ccf\backend\reports\60a23723-6e18-4305-aee5-f9b215da285e.pdf)
2026-07-21 15:04:25	QUERY_COPILOT	Asked Copilot: 'What is this investigation about?'. Response cited 5 artifacts.